



School of Computer Engineering

COURSE PLAN: THEORY COURSE

School Name	School of Computer Engineering				
Course Name	Software Engineering				
Course Code	ICT 3123	Semester	V	Core / PE / OE	Core
Name of the faculty	Dr. Raviraja Holla M				
No of contact hours/week:	L	T	P	C	
	3	0	0	3	

Mapping of Course Outcomes (COs)/Course Learning Outcomes (CLOs)

CO/ CLO	Statements	No. of Contact Hours	Marks	Program Outcomes (POs)	Learning Outcome s#	BL* (Recom mended)
1	Identify appropriate process models with the techniques to complete a small-scale analysis.	6	15	1,2	3	3 and above
2	Analyze the requirements of a project by translating it into appropriate models	11	33	3,4	4	3 and above
3	Analyze the architectural patterns with the design process.	5	14	7,11	5	3 and above
4	Apply suitable review and software testing techniques.	10	27	4,5	14	3 and above
5	Demonstrate the use of software configuration management and risk management concepts.	4	11	6,8,9,10	9	3 and above
	Total	36	100			

Applicable to IET Accredited Programs Only

*Preferred BL levels are above 3 (apply)

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2										2		
CO2			2	2								2		
CO3							2				2		2	
CO4				2	2									2

CO5						3		2	1	2			2	
Average Articulation Level	2	2	2	2	2	3	2	2	1	2	2	2	2	2

Module Matrix[#]

	LO1	LO2	LO3	LO4	LO5	LO6	LO7	LO8	LO9	LO10	LO11	LO12	LO13	LO14	LO15	LO16	LO17	LO18
CLO1	-	-	C	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CLO2	-	-	-	B	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CLO3	-	-	-	-	C	-	-	-	-	-	-	-	-	-	-	-	-	-
CLO4	-	-	-	-	-	-	-	-	-	-	-	-	-	C	-	-	-	-
CLO5	-	-	-	-	-	-	-	-	B	-	-	-	-	-	-	-	-	-

Applicable to IET Accredited Programs Only. Fill with B or C

Delivery and Assessment Plan of LOs[#]

Learning Outcome (LO) mapped to the course		Delivery and assessment Plan
LO	LO statement	
C3	Select and apply appropriate computational and analytical techniques to model complex problems, recognising the limitations of the techniques employed.	Lectures, directed reading, problem solving Assignment, Quiz, Examination
C4	Select and evaluate technical literature and other sources of information to address complex problems	Lectures, directed reading, problem solving Assignment, Quiz, Examination
C5	Design solutions for complex problems that meet a combination of societal, user, business and customer needs as appropriate. This will involve consideration of applicable health & safety, diversity, inclusion, cultural, societal, environmental and commercial matters, codes of practice and industry standards	Lectures, directed reading, problem solving Assignment, Quiz, Examination
C14	Discuss the role of quality management systems and continuous improvement in the context of complex problems.	Lectures, directed reading, problem solving Assignment, Quiz, Examination, Design Case studies
C9	Use a risk management process to identify, evaluate and mitigate risks (the effects of uncertainty) associated with a particular project or activity	Lectures, directed reading, problem solving Assignment, Quiz, Examination, Design Case studies

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The Assessment Plan shall be formulated as per the directives communicated by the Office of the AD(A) in every odd semester.

Component	Type	Max. Marks	Syllabus: Topics covered during	Schedule	Blooms taxonomy levels
IA 1	Quiz	5	July 24 2026 to Aug 10 2026	17 Aug 2026 – 29 Aug 2026	3 to 6
IA 2	Surprise test-Pen and Paper	10	August 10 2026 to September 3 2026	7 Sep 2026 – 21 Sep 2026	3 to 6
MID Semester examination		30	L1 – L18	Sept 25 2026 to Oct 01 2026	2 to 6
1A3	Quiz	5	Oct 3 2026 to Oct 15 2026	19 Oct 2026 – 31 Oct 2026	3 to 6
END Semester examination		50	L1 – L36	14 Nov – Dec 2026	3 to 6

Proration Guidelines for Missed Internal Assessment (IA) Components

If a student has missed one of the IA components and prorating is approved by the HOD/Dean/Associate Deans, the average of the remaining component scores may be calculated and proportionally scaled to match the total marks of the missed component.

If a student misses more than one IA component and the HOD/Dean/Associate Deans approves prorating for one, the average of the remaining component scores, with zero assigned to the other missed components (excluding the prorated one), can be scaled appropriately to match the total marks of the missed component.

Lesson Plan

L No	Topics	CO Addressed
0	<i>To address OBE philosophy towards NBA and IET accreditation followed by a discussion on Course Plan, Course Outcomes, CO-PO Mapping and Blooms Taxonomy, with prerequisites for the course, if any.</i>	--
1	Nature of the software, changing nature of the software, Software Engineering, Software Process	CO1
2	Software Engineering Practice, Software Engineering Myths	CO1
3	Tutorial 1 Software Process Models, Process activities, Coping with Change, Process Improvement	CO1
4	Agile methods, Agile development techniques, Agile project management	CO1
5	Scaling agile methods, Extreme Programming, Scrum	CO1
6	Functional and non-functional requirements, Requirement engineering process	CO1
7	Requirement Analysis, elicitation, Requirement specification, Requirement validation, Requirements Change	CO2
8	Scenario-based: Requirements Analysis, Use-case diagram	CO2
9	Activity diagram, Swimlane diagram	CO2
10	Class-based: Identifying analysis classes	CO2
11	Class diagram -relationships& dependencies	CO2
12	Class-based: Analysis packages, Behavior	CO2
13	Sequence Diagrams	CO2
14	State Diagrams	CO2
15	Web/Mobile Apps: Creating behavioral models: Sequence	CO2
16	Patterns for requirement modelling	CO2
17	Case Study: requirement modelling for web app or mobile app	CO2
18	Architectural Design: Design concepts	CO3
19	Architectural views, Architectural patterns, Architectural design	CO3
20	Application architectures -Architecture styles	CO3
21	Object-oriented design ,Pattern based architecture	CO3
22	Architectural Design: Implantation issues, Open-source development.	CO3
23	Cost impact of software defects, Defect amplification and removal	CO4
24	Software Testing Strategies: Test strategies for conventional software	CO4
25	Software Testing Strategies: Test strategies for object-oriented software	CO4
26	Debugging, Testing Conventional Applications- white box testing	CO4
27	Testing Conventional Applications- White box testing , black box testing	CO4
28	Black box testing Continued	CO4
29	Review metrics and their use, Review A formality spectrum	CO4
30	Informal reviews, Formal technical reviews, Postmortem evaluation	CO4
31	Case study: Testing web application or mobile application	CO4
32	Testing Object-Oriented Applications	CO4
33	Umbrella activities: Software Configuration Management	CO5

34	Risk Management	CO5
35	Software Maintenance	CO5
36	Software Re-engineering	CO5

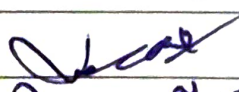
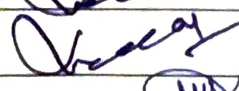
Self Directed Learning (SDL)

Sl. No.	TOPIC	No. of learning hours	CO
1	Study the evolution of Software Engineering and compare SDLC models through an NPTEL/MOOC lecture	4	CO1
2	Review case studies on Waterfall, Spiral, Agile, and DevOps software development models	5	CO1
3	Prepare a Software Requirement Specification (SRS) for a real-world application	5	CO2
4	Study IEEE SRS standards and analyse an industrial SRS document	4	CO2
5	Design Data Flow Diagrams (DFDs) and Structure Charts for a software system	5	CO3
6	Develop UML diagrams (Use Case, Class, Sequence, Activity, and State Chart) using StarUML/Visual Paradigm	6	CO3
7	Analyse object-oriented design case studies and identify suitable design patterns	5	CO3
8	Study principles of design patterns from research literature	4	CO3
9	Perform code review and static code analysis using coding standards and software quality guidelines	4	CO4
10	Study software testing techniques (Unit, Integration, System, Regression, Security, and Fuzz Testing) using case studies	4	CO4
11	Review recent research papers on Agile Software Engineering, DevOps, and AI-assisted Software Development	4	CO5
12	Complete an online NPTEL (https://onlinecourses.nptel.ac.in/e-learning/preview/noc25_cs108) /Coursera module (https://www.coursera.org/courses?query=software%20engineering) or Virtual Lab exercise on Software Engineering and prepare a summary report	4	CO5
Total Number of learning hours		54	

*The SDL topics may be from the Research Literature/ Case Studies NPTEL/Coursera/any MOOC-based material Virtual Lab, etc.

**One credit is equivalent to 30 hours of learning. (4 credits course =120 hrs learning ; 48+72=120 hrs of learning)

Faculty members teaching the course (if multiple sections exist):

Faculty Name	Section	Signature
Dr. Raviraja Holla M	A	
Dr. Raviraja Holla M	B	
Dr. Gaurang Mandal	C	
Dr. Lella Kranthi Kumar	D	

References:

Textbooks	1. Pressman R. S., <i>Software Engineering A practitioner's approach</i> , 8 th Edition, McGraw Hill, 2019.
	2. Sommerville Ian, <i>Software engineering</i> , 10 th Edition, Pearson Education, 2019.
	3. Booch G., Rumbaugh J., Jacobson I., <i>The Unified Modelling Language User Guide</i> , 2 nd Edition, Pearson, 2017.
	4. Booch G., Rumbaugh J., Jacobson I., <i>The Unified Modelling Language User Guide</i> , 2 nd Edition, Pearson, 2017.
	5. Rajib Mall, <i>Fundamentals of Software Engineering</i> , 5 th Edition, PHI Learning, 2018.
Self-Directed Learning	
Research Literature/ Case Studies	1. Sommerville, I., "Continuous Software Engineering", <i>IEEE Software</i> . 2. Recent research articles from IEEE Xplore, ACM Digital Library, and SpringerLink on Agile Software Development, DevOps, Software Testing, Design Patterns, and AI-assisted Software Engineering. 3. Case studies from Google Engineering, Microsoft Engineering, and ThoughtWorks Technology Radar.
NPTEL/Coursera/any MOOC-based material	NPTEL: <i>Software Engineering</i> by Prof. Rajib Mall (IIT Kharagpur) – https://archive.nptel.ac.in/noc/courses/noc21/SEM2/noc21-cs65/ NPTEL: <i>Software Testing</i> by Prof. Rajib Mall (IIT Kharagpur) – https://archive.nptel.ac.in/ (Software Testing course archive) SWAYAM: Latest NPTEL offering of Software Engineering by Prof. Rajib Mall.
Virtual Labs	Virtual Labs (Ministry of Education, Government of India) – https://www.vlab.co.in/ Relevant labs: Software Engineering Lab, Software Testing Lab, UML Modelling, and Requirements Engineering (where available). Students may also use StarUML, Visual Paradigm Community Edition, or draw.io for UML modelling practice.

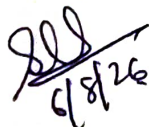
Submitted by: Dr. Raviraja Holla M



(Signature of the faculty)

Date: 31.07.2006

Approved by:



(Signature of the Dean)

